Hypothesis testing. A hypothesis test starts with a null hypothesis, usually stating no effect or no difference. The p-value is the probability of observing data at least as extreme as the sample if the null hypothesis were true; it is not the probability that the null hypothesis is true. A result is called statistically significant when the p-value is below the chosen significance level alpha, commonly 0.05. A type I error rejects a true null hypothesis and occurs with probability alpha. A type II error fails to reject a false null hypothesis; the power of a test is one minus the probability of a type II error.

Confidence intervals and the central limit theorem. The central limit theorem states that the distribution of the sample mean approaches a normal distribution as the sample size grows, whatever the shape of the population distribution, provided the variance is finite. A 95 percent confidence interval is built by a procedure that captures the true parameter in 95 percent of repeated samples. It does not mean there is a 95 percent probability that a specific computed interval contains the parameter. Larger samples produce narrower intervals because the standard error of the mean equals the standard deviation divided by the square root of n.

ANOVA and model comparison. One-way ANOVA tests whether the means of three or more groups are equal by comparing between-group variance with within-group variance through the F-statistic. A large F-statistic with a small p-value suggests at least one group mean differs. ANOVA does not say which groups differ, so a post hoc test such as Tukey's HSD is used. For comparing regression models, AIC and BIC reward goodness of fit but penalise the number of parameters; BIC applies a heavier penalty than AIC when the sample is large. RMSE is the square root of the mean squared prediction error and is in the same units as the response.